

SEAT NO. _____

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
SECOND YEAR(Computer Science)
SPRING SEMESTER EXAMINATIONS 2026

Time : 3 Hours

Batch 2024

Dated : 06-JUN-26

Max Marks : 60

Database Management Systems - CT-261

Instructions:

- Attempt all questions
- Avoid unnecessary details
- Draw the diagrams with pencils
- Neat and clean work shall attract more marks

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Question#1

(CLO-1)

- a) **Describe** the reasons of having three levels of abstraction in ANSI-SPARC architecture of DBMS. **(4 marks)**
- b) **Discuss** the term 'client-server architecture' and what are the advantages of this approach? Compare the client-server architecture with other two architectures. **(6 marks)**
- c) **Discuss** the properties of a relation. **(2 marks)**

Question#2

(CLO-1)

- a) **Discuss** an indexing strategy for a university database containing Student_ID, Name, Department, and CGPA. Explain suitable indexing types for different attributes with examples, and justify how your design improves query performance. **(3 marks)**
- b) Explain different states of transactions with Diagram. **(3 marks)**
- c) **Describe** timestamp? How do timestamp-based protocols for concurrency control differ from locking-based protocols? **(3 marks)**
- d) **Demonstrate** the problem of Uncommitted Data with example due to the lack of Concurrency Control. **(3 marks)**

Question#3

(CLO-2)

- c) **Design** approaches for securing DBMSs on the Web against SQL injection attacks. Illustrate a simple SQL injection example and explain how security mechanisms help prevent unauthorized database access. **(6 marks)**
- a) Consider the following relational database schema consisting of the four relation schemas: **(6 marks)**
 - passenger (pid, pname, pgender, pcity)
 - agency (aid, aname, acity)
 - flight (fid, fdate, time, src, dest)
 - booking (pid, aid, fid, fdate)

Months

Generate the relational algebra expressions for the following queries:

- i. Get the details about all flights from Karachi to New York.
- ii. Find the passenger names for passengers who have bookings on at least one flight.
- iii. Get the details of flights that are scheduled on either of the dates 01/12/2024 or 02/12/2024 or both at 16:00 hours.
- iv. Find the agency names for agencies that do not have any bookings for passenger with id 123.

Question#4

(CLO-2)

An organization is developing an integrated database system to manage its employees, departments, projects, and library resources. Each employee is uniquely identified by an employee number and has attributes such as employee name and employee address. Each department is identified by a department number and also has a department name and department location. Every employee works in exactly one department, and each department must have one or more employees working in it; there is no department without employees, and no employee exists without being assigned to a department.

Employees are further categorized into three mutually exclusive types: hourly employees, salaried employees, and consultant employees. Each hourly employee has an hourly rate, each salaried employee has a monthly salary, and each consultant employee has a billing rate. Every employee belongs to exactly one of these categories.

The organization also manages projects. Each project has a project number, title, and duration. Projects are classified into three types: research projects, development projects, and educational projects. A research project stores research grant information, a development project stores the funding organization, and an educational project stores the name of the educational institute. Employees may work on one or more projects, and each project may have one or more employees assigned to it. For each assignment, the system stores the assignment date and assignment status.

The organization maintains a library system that contains books. The library has attributes such as location, start time, and end time. Each library contains one or more books, and each book belongs to exactly one library. Each book is identified by a book number and has attributes such as title, authors, publisher, and edition. Books are further classified into IT books, management books, and humanities books. IT books store information about supported tools, management books store organizational case studies, and humanities books store art type information. Employees may borrow books from the library. A book may be issued to only one employee at a time, although an employee may borrow multiple books. Some employees may not borrow any books, and some books may remain unissued. For each book issue, the system records issue date, due date, and fine amount if applicable. Construct an Enhanced Entity-Relationship (EER) diagram for the above system.

1. Identify entities, including superclasses and subclasses (if any).
2. Identify relationships among entities and their participation.
3. Identify attributes of entities and relationships, including primary keys.
4. Identify constraints, including cardinality, participation, and specialization (disjoint/overlapping, total/partial). State assumptions if needed.

(12 marks)

Question#5**(CLO-2)**

- a) Given the following Invoice data (Pine Valley Furniture Company). Construct the functional dependencies and normalize it to 3NF. State any assumptions you make about the data shown below: **(8 marks)**

<u>OrderID</u>	<u>Order Date</u>	<u>Customer ID</u>	<u>Customer Name</u>	<u>Customer Address</u>	<u>ProductID</u>	<u>Product Description</u>	<u>Product Finish</u>	<u>Product StandardPrice</u>	<u>Ordered Quantity</u>
1006	10/24/2024	2	Value Furniture	Plano, TX	7	Dining Table	Natural Ash	800.00	2
					5	Writer's Desk	Cherry	325.00	2
					4	Entertainment Center	Natural Maple	650.00	1
1007	10/25/2024	6	Furniture Gallery	Boulder, CO	11	4-Dr Dresser	Oak	500.00	4
					4	Entertainment Center	Natural Maple	650.00	3

- b) The given table is susceptible to update anomalies. **Explore** examples of insertion, deletion, and modification anomalies. Primary key attributes are underlined. **(4 marks)**

<u>StdSSN</u>	<u>StdCity</u>	<u>StdClass</u>	<u>OfferNo</u>	<u>OffTerm</u>	<u>OffYear</u>	<u>EnrGrade</u>	<u>CourseNo</u>	<u>CrsDesc</u>
S1	SEATTLE	JUN	O1	FALL	2006	3.5	C1	DB
S1	SEATTLE	JUN	O2	FALL	2006	3.3	C2	VB
S2	BOTHELL	JUN	O3	SPRING	2007	3.1	C3	OO
S2	BOTHELL	JUN	O2	FALL	2006	3.4	C2	VB